

Skills Summary

Powers of Ten and Place-Value Relationships

Use this reference while completing your worksheet or when studying.

One rule runs the whole system. Each place is worth 10 times the place to its right and $\frac{1}{10}$ of the place to its left.

hundred th.	ten th.	thousands	hundreds	tens	ones	tenths	hundredths
10^5	10^4	10^3	10^2	10^1	10^0	$\frac{1}{10}$	$\frac{1}{100}$

An exponent on 10 counts *places*, and for whole numbers it also counts *zeros*: $10^4 = 10000$.

1. Comparing two digit values

Compare **values**, never digits. Write what each digit is worth, then divide the larger value by the smaller. Places one apart give 10; two apart give 100; three apart give 1000.

Example: In 770, how many times as much is the first 7 worth as the second 7?

Step 1. The two digits are the same, so only their places can differ — write each one's value before comparing anything.

Step 2. The first 7 is worth 700 and the second is worth 70, and $700 \div 70$ gives the comparison.

10 times as much (one place apart)

Try it: In 90900, how many times as much is the 9 in the ten-thousands place worth as the 9 in the hundreds place?

check: 100

2. Powers of ten in exponent form

10^n means 10 used as a factor n times: $10^1 = 10$, $10^2 = 100$, $10^3 = 1000$, and $10^0 = 1$.

A number that ends in zeros can be written as a digit times a power of ten: $700 = 7 \times 10^2$. Expanded form uses one term per non-empty place.

Example: Write the value of 10^5 .

Step 1. The exponent is a count of factors of ten, so it is also a count of zeros for a whole-number power — no long multiplication is needed.

Step 2. Five factors of ten means a 1 followed by five zeros.

$10^5 = 100000$

Try it: Write 8×10^3 in standard form.

check: 8000

3. Shifting digits by 10, 100 and 1000

Multiplying by 10^n shifts every digit n places **left**; dividing by 10^n shifts every digit n places **right**. Empty whole-number places take 0 as a placeholder.

Say “the digits move”, not “add a zero” — the zero rule fails the moment a decimal appears.

Example: Find 2.4×100 .

Step 1. $100 = 10^2$, so this is a two-place shift to the left rather than a multiplication to work out.

Step 2. The 2 moves from ones to hundreds and the 4 moves from tenths to tens, leaving the ones place empty, so it takes a 0.

$$2.4 \times 100 = \mathbf{240}$$

Try it: Find $65 \div 10$.

check: 6.5

4. Rounding to a place to estimate

Rounding to a place *is* rounding to a power of ten: nearest ten = 10^1 , nearest hundred = 10^2 , nearest thousand = 10^3 .

Find the place you were asked for, look at the **one digit to its right**, and round up when that digit is 5 or more. Every place to the right becomes 0. Then compute with the rounded numbers.

Example: Estimate $382 + 419$ to the nearest hundred.

Step 1. An estimate is exact arithmetic on rounded numbers, so round each addend first and only then add.

Step 2. 382: the tens digit is 8, so round up to 400. 419: the tens digit is 1, so round down to 400. Now $400 + 400$.

$$\mathbf{800}$$

Try it: Estimate $5170 - 2840$ to the nearest thousand.

check: 2000

(!) **Watch out:** count how many places apart two digits are before you answer “ten times” — two places apart is 100 times. And round by the place you were *asked* for: to round to the nearest hundred, read the tens digit, not the ones digit.